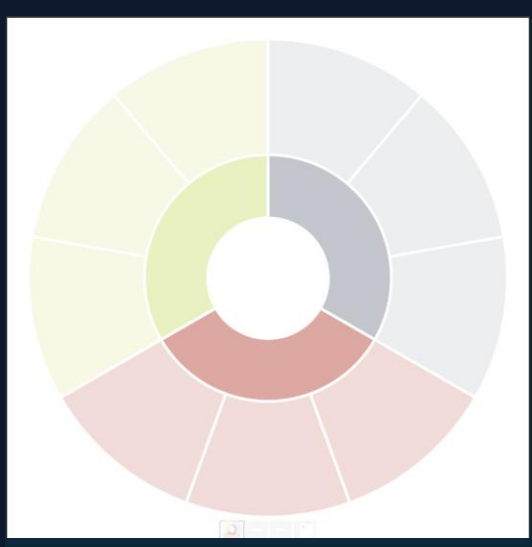




CHaRM-AI

Institution-Specific LLM Based Intelligent Question Answering System

Project Author : Ceyhun Kirca

Supervisor : Prof. Dr. Mine Elif Karsligil



PROBLEM & CURRENT SITUATION

Industry Problem: Growing volumes of institutional and technical documents make efficient access to relevant information increasingly difficult.

Current Challenges: Keyword-based search methods often misses contextual meaning.

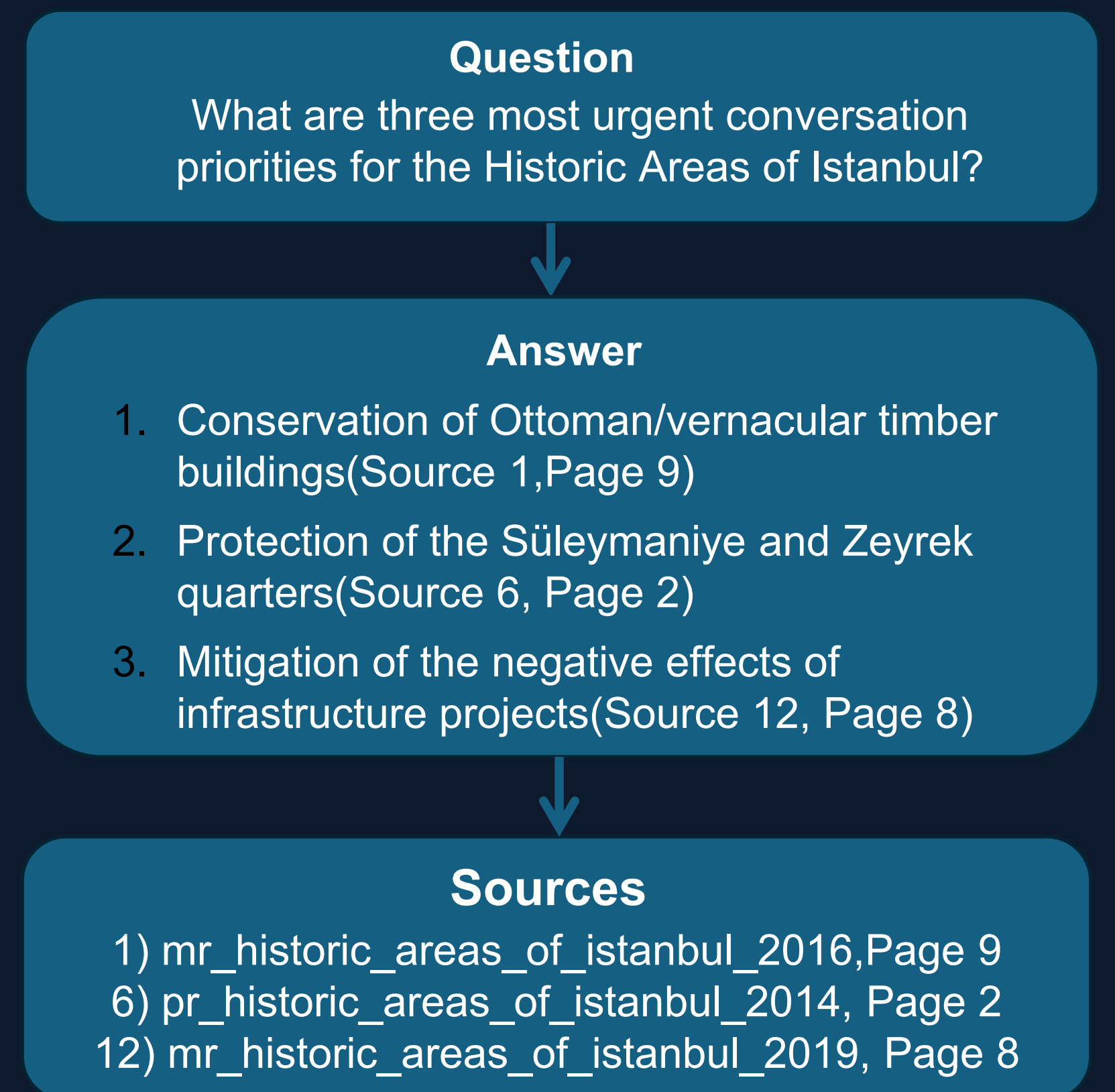
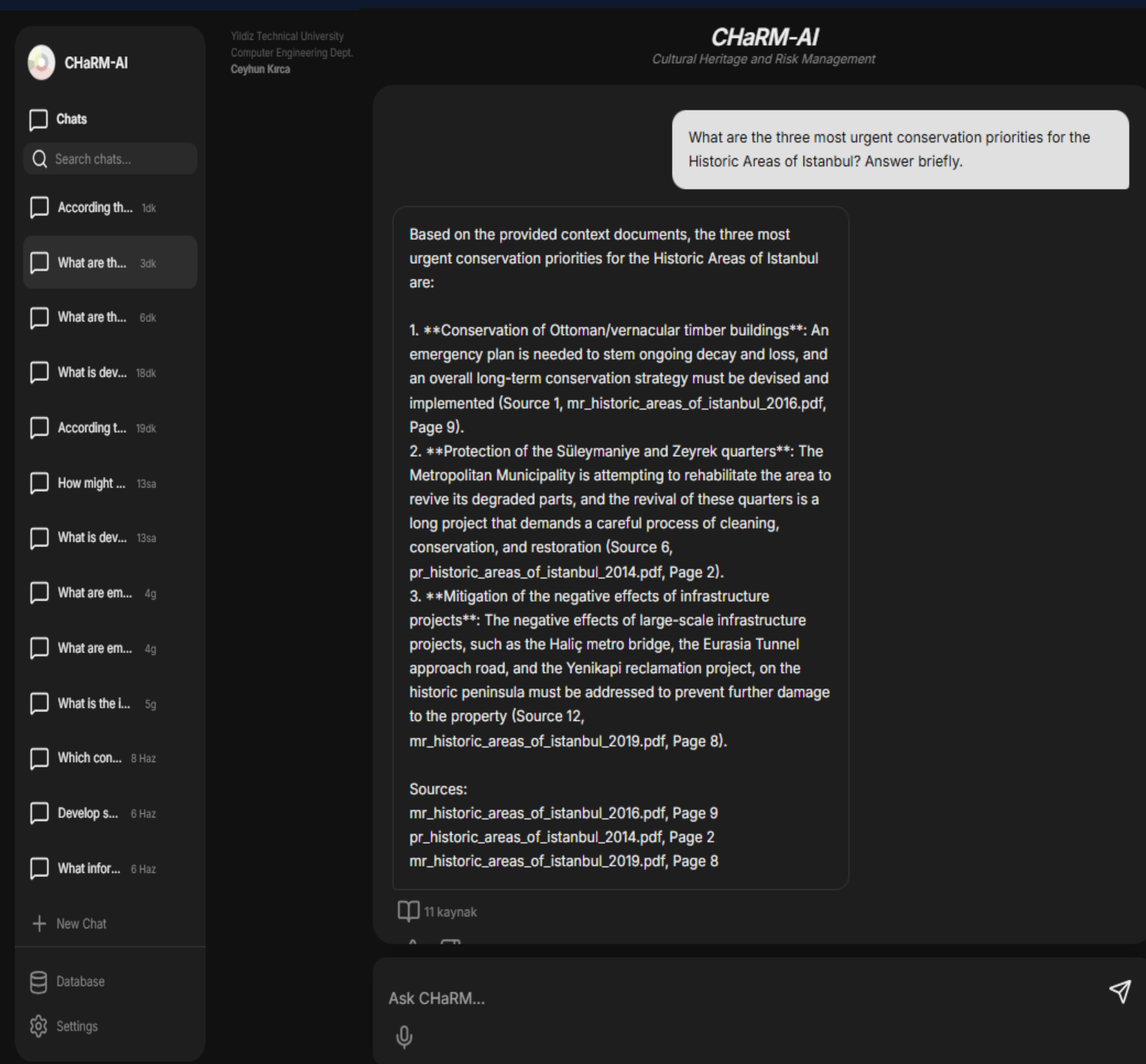
Objective and Need : Develop a reliable question answering system that retrieves relevant content and provides source-supported responses.

PROPOSED SOLUTION & INNOVATION

Core Approach: A RAG-based question answering system combining semantic retrieval and response generation.

Innovation: Generates answers using retrieved document context instead of relying only on model knowledge.

Added Value : Faster information access, reduced manual search effort, and reliable source-supported responses.



KEY FINDINGS & CONCLUSION

93.1% Weighted RAGAS Score

- Improved semantic retrieval with two-stage retrieval and reranking
- Reduced hallucinations and increased response reliability
- Fast inference and source-supported answers

CHaRM-AI transformed unstructured cultural heritage documents into an accurate and searchable knowledge base.

